

FinSight AI

Product Requirements Document (PRD) & System Design Document (SDD)

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1. Executive Summary

FinSight AI is an AI-powered Financial Intelligence Platform that combines personal finance management, investment research, portfolio analytics, and agentic AI. The platform ingests private financial documents and public market information, enabling users to receive citation-backed, explainable insights.

Objectives

Build a production-grade AI system demonstrating hybrid RAG, multi-agent orchestration, observability, evaluation, asynchronous processing, and scalable backend architecture.

Success Metrics

- End-to-end query latency <2.5s (cached)
- Retrieval precision >85%
- Faithfulness >90% on evaluation dataset
- 95% successful document ingestion
- Modular deployment with Docker.

2. User Personas

Working Professional

Track expenses, analyze salary, budget planning, affordability analysis.

Retail Investor

Research companies, summarize earnings, compare competitors, analyze portfolio.

Power User

Deep research, long-context document analysis, downloadable reports.

3. Product Modules

Personal Finance

Statement ingestion, categorization, budgeting, subscriptions, cash-flow, financial health.

Market Intelligence

Annual reports, earnings calls, news, financial ratios, AI Q&A.;

Portfolio Intelligence

Asset allocation, diversification, risk analysis, watchlists.

Financial Copilot

Natural language reasoning over personal + public financial knowledge.

4. Functional Requirements

Authentication: JWT, refresh tokens, RBAC.

Document Management: Upload PDF/CSV/XLSX/DOCX, OCR, metadata extraction.

Analytics: Expense trends, income, savings, goals, investment metrics.

Research: Citation-backed RAG, company comparison, report generation.

Chat: Streaming responses, conversation history, source attribution.

5. Non-Functional Requirements

Availability: 99.9%. Security: HTTPS, encryption at rest, audit logs, secrets management.

Scalability: stateless services with horizontal scaling. Observability: tracing, metrics, logs.

Performance: async processing, Redis caching.

6. High-Level Architecture

Client (Next.js) | API Gateway (FastAPI) | Auth | User | Document | Portfolio | Analytics | AI Services
| RabbitMQ/Celery | OCR -> Parsing -> Chunking -> Embeddings | Hybrid Retrieval (BM25 +
pgvector + Reranker) | LangGraph Orchestrator | LLM Gateway (OpenAI/Ollama/Claude/Gemini) |
PostgreSQL | Redis | Object Storage | LangSmith | Phoenix | Prometheus | Grafana

7. AI & Agent Architecture

Planner Agent

Classifies intent and routes tasks.

Finance Agent

Expense, income, affordability, forecasting.

Research Agent

Retrieves public financial knowledge and validates sources.

Portfolio Agent

Risk, allocation, diversification.

Reasoning Agent

Combines outputs into a coherent answer.

Evaluation Agent

Runs automatic quality checks and metrics.

8. RAG Pipeline

Upload → OCR → Cleaning → Metadata → Chunking → Embeddings → pgvector + BM25 → Hybrid Retrieval → Cross Encoder Reranker → Context Builder → LLM → Citation Validator → Response

Chunk strategy: semantic chunking with overlap. Retrieval: reciprocal rank fusion. Re-ranking: Cross Encoder. Cache: Redis.

9. Database Design

Users, Documents, DocumentChunks, Transactions, Portfolios, ConversationHistory, Evaluations, PromptVersions, AuditLogs.

Indexes on embeddings, transaction dates, ticker, user_id. Partition transaction tables by month.

10. API Design

POST /auth/login POST /documents/upload POST /chat GET /dashboard GET /portfolio POST /portfolio/analyze POST /research/company GET /reports/{id}

Responses are JSON with streaming support via Server-Sent Events.

11. Security

JWT authentication, RBAC, encrypted secrets, signed URLs for uploads, prompt injection protection, malicious document scanning, PII masking, rate limiting.

12. Observability & Evaluation

Metrics: p50/p95 latency, retrieval latency, embedding latency, LLM latency, token usage, cost, queue depth, cache hit rate.

Evaluation: RAGAS, DeepEval, hallucination detection, faithfulness, context precision, regression suite in CI/CD.

13. Deployment

Local development with Docker Compose. Production via Kubernetes (optional), Nginx ingress, GitHub Actions CI/CD, PostgreSQL, Redis, object storage, monitoring stack.

14. Roadmap

Sprint 1: Infrastructure, Auth, CI/CD.

Sprint 2: Document ingestion & OCR.

Sprint 3: Hybrid RAG.

Sprint 4: Personal Finance.

Sprint 5: Portfolio Intelligence.

Sprint 6: Market Intelligence.

Sprint 7: Multi-Agent AI.

Sprint 8: Observability & Evaluation.

Sprint 9: Deployment & Documentation.

15. Future Enhancements

Voice interface, mobile app, OCR for handwritten statements, multilingual support, advanced forecasting, financial planning simulations, advisor mode.

16. Resume Highlights

Production-grade AI platform with hybrid RAG, LangGraph multi-agent orchestration, asynchronous ingestion pipeline, evaluation framework, observability, Dockerized deployment, and local/cloud LLM benchmarking.